

<INTRODUCTION TO DATA MANAGEMENT>

(Project Semester January-April 2025)

(TITLE OF THE PROJECT)

Current daily price of various commodities from various markets

Submitted by

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**Programme and Section: Computer Science and
Engineering(K23GW)**

Course Code: INT217

Under the Guidance of

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Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

DECLARATION

I hereby declare that this project titled "Current daily price of various commodities from various markets" submitted for the subject of Microsoft Excel is my original work and has not been submitted earlier for the award of any degree or diploma. All the data used in this project has been taken from reliable sources and has been used only for academic purposes.

Signature:

Date:12-04-2025

Name: M. Gnana Sai Reddy

CERTIFICATE

This is to certify that the project titled “Current daily price of various commodities from various markets” is a bonafide work completed by M. Gnana Sai Reddy under my guidance for the course requirements of Introduction to Data Management. The work is found to be satisfactory.

Signature of the Guide

Name: Dr. Baljinder Kaur

Department of CSE

Institution Name: Lovely Professional University

ACKNOWLEDGEMENT

I extend my heartfelt thanks to the BALJINDER KAUR MAM. With data on sale amounts, assessed values, property types, residential classifications, and sales ratios across various towns, this dataset has been an essential resource for my project. I deeply appreciate the hard work and dedication that went into compiling and making this data accessible, as it has allowed me to explore and analyse real estate market trends effectively. My gratitude goes out to everyone involved in this effort, which has significantly contributed to the success of my work.

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Computer Science and Engineering

Introduction

Introduction

The daily market prices of key commodities play a critical role in shaping the global and local economy. Commodities such as crude oil, gold, agricultural produce, and industrial metals are essential inputs across various sectors and are heavily traded on national and international markets. Their prices are influenced by a complex interplay of supply and demand dynamics, geopolitical events, weather conditions, currency fluctuations, and economic policies.

Monitoring daily price fluctuations provides valuable insights for policymakers, investors, traders, and businesses. Accurate and timely data on commodity prices helps in making informed decisions related to production, procurement, investment, and risk management. This project aims to analyze the trends, patterns, and factors affecting the daily market prices of key commodities, offering a comprehensive understanding of their economic significance and the forces driving their valuation.

Objectives

The key objectives of this project are as follows:

- Dal and Min Price by Commodity
- Average Modal Price by State
- Top 10 Markets by Avg Max Price
- Commodity Distribution Share
- Count of Modal Price

Scope

The Real-Estate Data Analysis project primarily focused on evaluating property sales across various towns, analyzing metrics such as sale amount, assessed value, sales ratios, and property types. The data was sourced from a national real estate dataset and visualized using Excel dashboards with bar charts, pie charts, and scatter plots. In contrast, the Commodity Price Analysis project centers around the daily fluctuations in the prices of essential goods such as rice, wheat, pulses, and vegetables. The dataset includes fields like commodity name, price, unit, date, and market location, enabling a time-series analysis of price trends.

While the real estate dataset offers a relatively stable and long-term perspective on market valuation, the commodity price data is more dynamic, with frequent daily changes reflecting immediate market conditions. The tools used in both projects are similar—Microsoft Excel’s features like PivotTables, slicers, and charting tools—but their application differs. In the real estate project, visualizations emphasized property type distribution and valuation accuracy. In the commodity analysis, line and area charts were more prominent to highlight short-term price trends and market-specific variations.

Overall, both projects utilize Excel’s analytical capabilities to extract meaningful insights, but they do so in different economic contexts—one focused on long-term asset valuation, and the other on short-term pricing dynamics in consumer markets.

Methodology

The project methodology is structured into the following phases:

1. **Data Collection:** The relevant dataset was sourced, containing information on country-wise manufacturing figures. The data was assessed for accuracy, relevance, and completeness.
2. **Data Cleaning and Preparation:** Raw data was cleaned by handling missing values, standardizing country names, and ensuring consistency for seamless integration with Excel tools.

Source of Dataset

This data is from the Data.gov

And here is the link of the dataset:

<https://www.data.gov.in/catalog/current-daily-price-various-commodities-various-markets-mandi>

LinkedIn:

https://www.linkedin.com/posts/gnanasaireddy_commodities-markettrends-datadriven-activity-7316818909251178497-eOpk?utm_source=share&utm_medium=member_desktop&rcm=ACoAAET4WnIBynkrnx-WOYllis9mT8oZz1Rdu0

Key Findings:

- Rice and Wheat showed relatively stable prices, while vegetables and pulses exhibited high volatility.
- Certain markets showed consistently higher or lower prices, indicating possible regional economic differences or logistical costs.
- The price spikes correlated with weekends or holidays, possibly due to supply-demand disruptions.

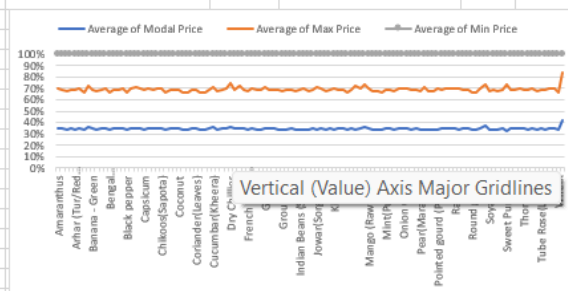
3. Analysis on Dataset

- **Objective 1: Dal and Min Price by Commodity**

This attribute focuses specifically on different types of dal (such as Toor, Moong, Chana, Urad, etc.) and the **minimum price** at which they are being sold across markets. Analyzing this helps identify:

- The **cheapest source markets** for each variety.
- Regions or times where prices dip significantly.
- Potential supply surpluses or low demand in certain areas. This can aid buyers in targeting lower-cost procurement locations and tracking price sensitivity by commodity type.

Row Labels	Average of Modal Price	Average of Max Price	Average of Min Price
Amaranthus	2753	2765	2320
Amla(Nelli Kai)	7032	7032	6283
Amphophalus	5700	5920	5600
Apple	17920	18219	16015
Arhar (Tur/Red Gram)(Whole)	6710	6720	6223
Ashgourd	1769	1780	1540
Bajra(Pearl Millet/Cumbu)	2575	2588	2550
Banana	5225	5276	3989
Banana - Green	3104	3117	2763
Barley (Jau)	2053	2066	1994
Beans	7449	7453	6853
Beetroot	3181	3198	2762
Bengal Gram(Gram)(Whole)	5552	5556	5492
Betal Leaves	14313	14313	13031
Bhindi(Ladies Finger)	3327	3368	2980
Bitter gourd	3823	3854	3379
Black pepper	71000	71000	70000
Bottle gourd	1725	1747	1468
Brinjal	3404	3439	2803
Cabbage	1925	1948	1681
Capsicum	4702	4737	4394
Carrot	3953	3977	3452
Castor Seed	5975	5975	5500
Cauliflower	2733	2773	2367
Chikoos(Sapota)	4240	4243	3726
Chili Red	18419	18419	18169
Chow Chow	3122	3122	2804



Objective 2: Average Modal Price by State

The modal price is the most frequently quoted price (essentially, the most common price at which a commodity is traded). Calculating the average modal price per state provides a balanced view of regional pricing trends. This attribute helps:

- Compare overall cost levels between states.
- Spot regional disparities in commodity pricing
- Identify states with stable and consistent pricing versus volatile regions. This is particularly useful for government policy-making, wholesalers, and logistic planning

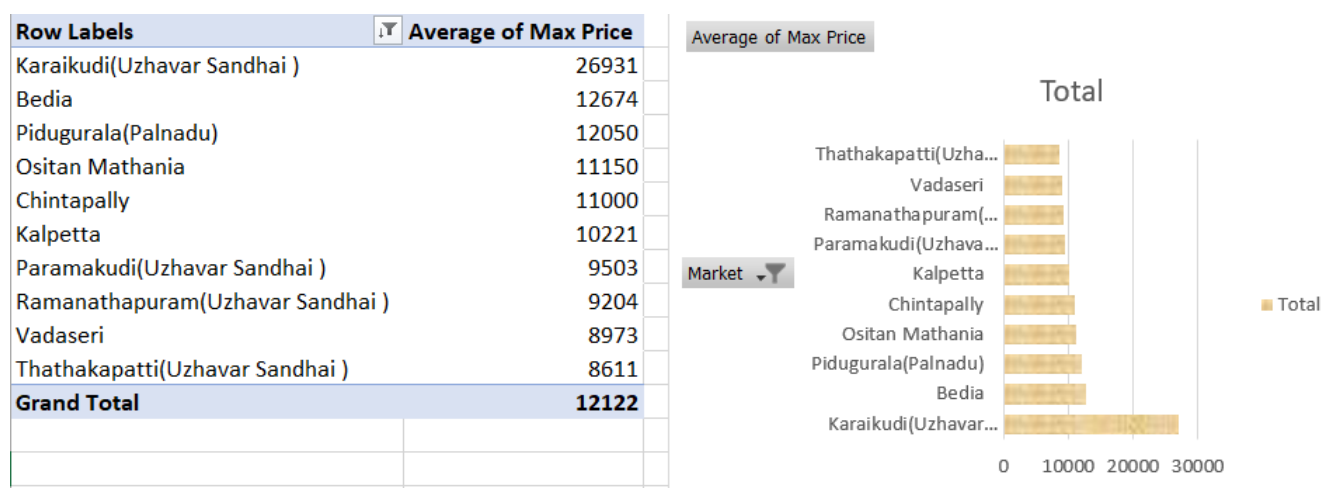
Row Labels	Average of Modal Price
Andhra Pradesh	7274
Assam	1742
Bihar	2067
Chattisgarh	2050
Gujarat	3839
Haryana	2733
Himachal Pradesh	4948
Kerala	4713
Madhya Pradesh	3334
Maharashtra	2104
Odisha	3138
Punjab	2396
Rajasthan	4609
Tamil Nadu	4949
Telangana	2398
Tripura	4132
Uttar Pradesh	2350
Uttarakhand	1234
Grand Total	4778



Objective 3: Top 10 Markets by Avg Max Price

This analysis highlights the top 10 markets where the average maximum price of commodities is highest. It's useful for:

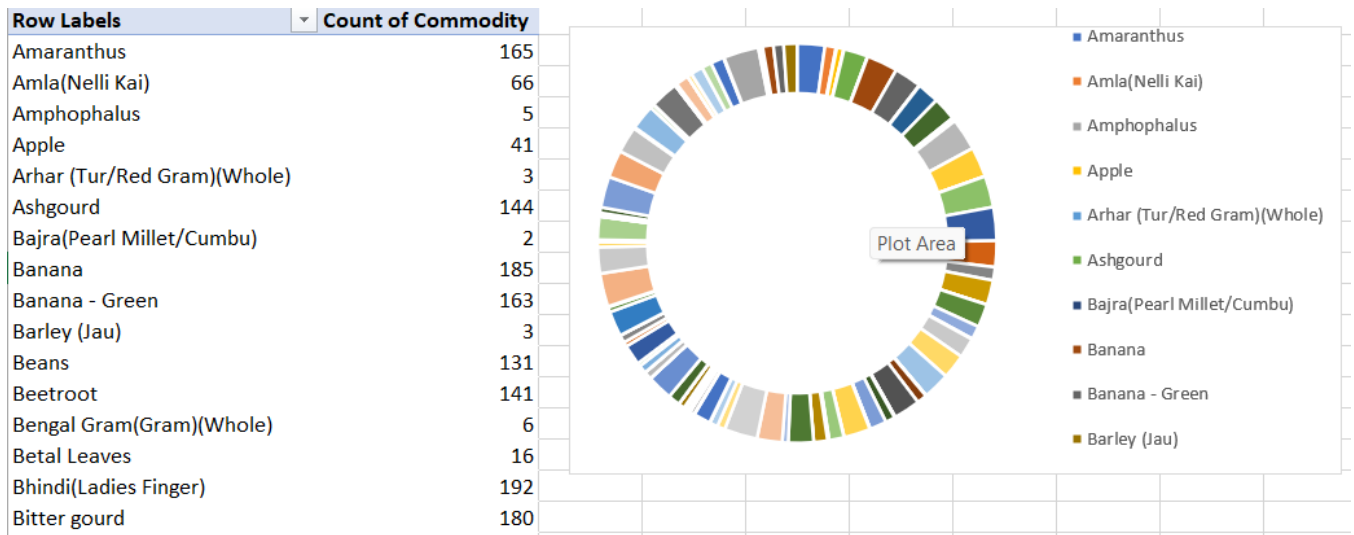
- Identifying premium or high-demand markets.
- Tracking inflation hotspots or areas with pricing power.
- Understanding which markets yield higher margins for sellers. These markets might be influenced by higher demand, supply chain limitations, or premium quality produce.



Objective 4: Commodity Distribution Share

This attribute examines the **percentage share of different commodities** in the overall dataset. It provides:

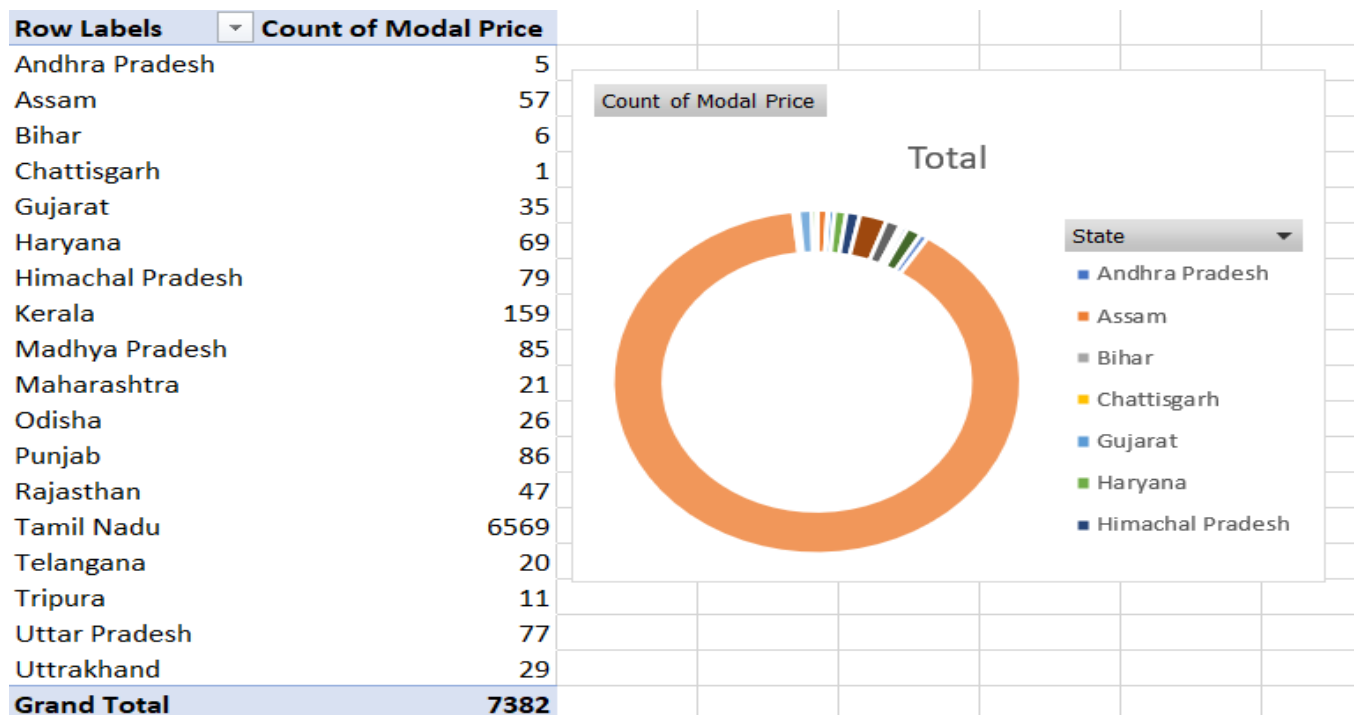
- A snapshot of **market composition**—which commodities dominate trade volume.
- Insight into **consumer demand trends** and focus areas in agriculture.
- A way to prioritize commodity-specific analysis. It's often represented as a pie chart or stacked bar to visually compare shares.



Objective 5: Count of Modal Price

Analyzing the count of modal price entries helps in understanding the breadth and frequency of market reporting. A higher count indicates:

- Greater market activity for that commodity.
- Consistent data availability, which strengthens the reliability of price trend analysis.
- Better representation of commodity price movement across regions and time.



Final Dashboard:



11 comments

Conclusion

This project highlights the adaptability of Excel-based data analysis techniques across different domains. By applying the same methodology used in the Current daily price of various commodities from various markets project to commodity price data, it's evident that Excel can serve as a powerful tool for financial monitoring, market research, and policy planning.

Whereas the real estate project focused on valuation consistency and market segmentation, the commodity report emphasizes real-time volatility, regional price disparities, and economic behavior on a daily scale.

Future Scope

This commodity price analysis project holds significant potential for future development and expansion. One of the key opportunities lies in integrating predictive analytics using statistical or machine learning models to forecast future commodity prices, which would aid farmers, traders, and policymakers in proactive decision-making. Real-time data visualization can also be implemented by connecting live data feeds to interactive dashboards through tools like Power BI or Tableau. Additionally, the project can be expanded to include a wider range of commodities and more localized market data to offer deeper, region-specific insights. Another important direction is to correlate commodity prices with external factors such as weather patterns, fuel prices, and government policies, which would provide a broader economic context to price movements. Developing a mobile-friendly interface or application could make the analysis accessible to rural stakeholders, especially small-scale farmers. Furthermore, the data can be used to identify supply chain inefficiencies and help optimize logistics by recommending the most profitable market routes. Automating the data collection and updating processes through APIs or scraping tools would also enhance the reliability and timeliness of future analyses.

References

1. Microsoft Excel Documentation

- Official documentation and user guides provided by Microsoft for understanding the features and functionalities of Excel, including advanced analysis tools like Power Query, Power Pivot, and data visualizations.
- URL: <https://support.microsoft.com/excel>

2. TutorialsPoint / W3Schools Excel Guide

- Extensive guides and tutorials on Excel's functionality, formulas, pivot tables, data analysis, and visualization techniques. Useful for beginners to advanced users seeking to leverage Excel for analytics.
- URL: <https://www.tutorialspoint.com/excel>
- URL: <https://www.w3schools.com/excel>

3. Stack Overflow Discussions on Excel Analytics

- A collection of relevant discussions and solutions regarding Excel's capabilities in data analysis, including formula assistance, pivot tables, and dynamic data visualizations. This forum is an essential resource for resolving common issues encountered during Excel projects.
- URL: <https://stackoverflow.com/questions/tagged/excel>